In [1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import confusion_matrix, accuracy_score, precision_score, reca
```

In [2]:
```
df=pd.read_csv(r"C:\Users\System05\Desktop\13317\csv_dataset\Social_Network_Ads.csv
print(df.head())
```

```
   Age  EstimatedSalary  Purchased
0   19            19000          0
1   35            20000          0
2   26            43000          0
3   27            57000          0
4   19            76000          0
```

In [3]:
```
print(df.isnull().sum())
```

```
Age                0
EstimatedSalary    0
Purchased          0
dtype: int64
```

In [4]:
```
print(df.cov())
```

```
                          Age  EstimatedSalary    Purchased
Age                109.890702     5.548738e+04     3.131165
EstimatedSalary  55487.380952     1.162603e+09  5924.367168
Purchased            3.131165     5.924367e+03     0.230269
```

In [5]:
```
X = df[['Age', 'EstimatedSalary']].values
Y = df['Purchased'].values
```

In [6]:
```
xtrain, xtest, ytrain, ytest = train_test_split( X, Y, test_size=0.25, random_state
```

In [7]:
```
sc = StandardScaler()
```

In [8]:
```
xtrain = sc.fit_transform(xtrain)
xtest = sc.transform(xtest)
```

In [9]:
```
logreg = LogisticRegression()

logreg.fit(xtrain, ytrain)
```

Out[9]:
```
▾ LogisticRegression
LogisticRegression()
```

In [11]:
```
y_pred = logreg.predict(xtest)
```

In [12]:
```
cm = confusion_matrix(ytest, y_pred)
```

In [13]:
```
TN, FP, FN, TP = cm.ravel()
```

In [15]:
```
print(cm)
```

```
[[65  3]
 [ 8 24]]
```

In [22]:
```
print(f"True Positives (TP): {TP}")
print(f"False Positives (FP): {FP}")
print(f"True Negatives (TN): {TN}")
print(f"False Negatives (FN): {FN}")
```

```
True Positives (TP): 24
False Positives (FP): 3
True Negatives (TN): 65
False Negatives (FN): 8
```

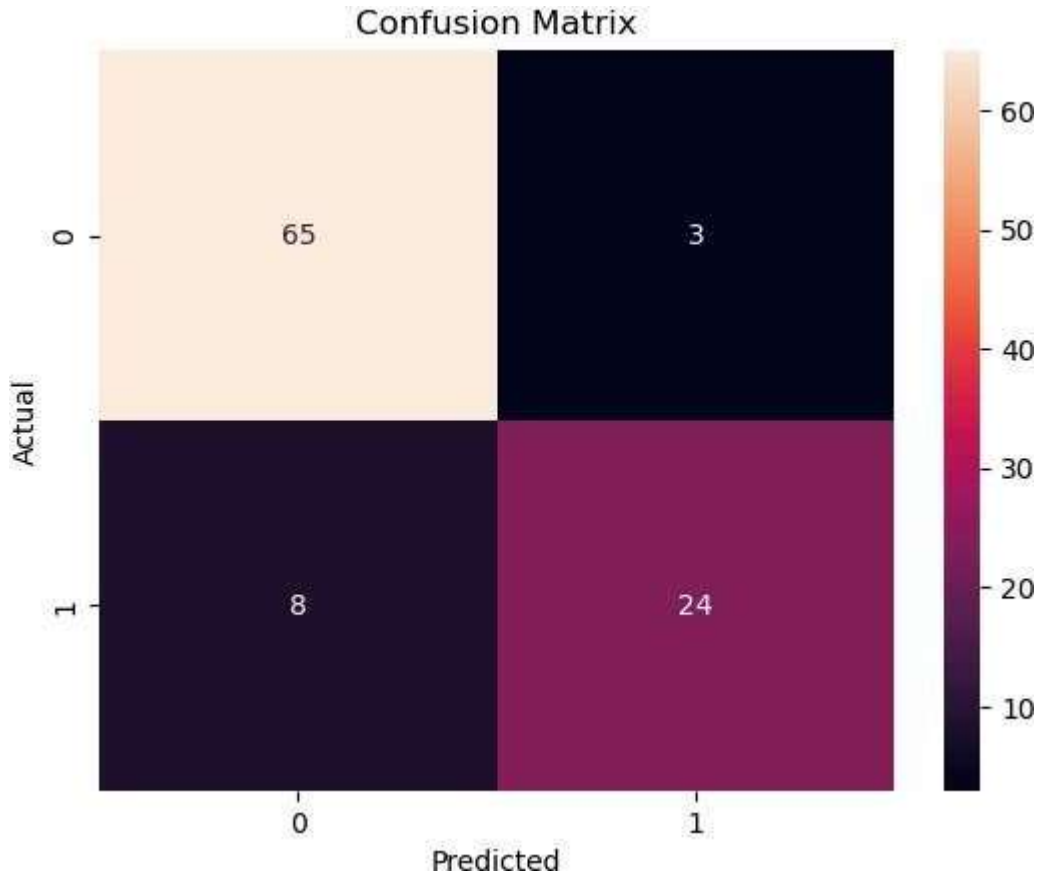
In [17]:
```
plt.figure()
```

Out[17]:
```
<Figure size 640x480 with 0 Axes>
```

```
<Figure size 640x480 with 0 Axes>
```

In [21]:
```
sns.heatmap(cm, annot=True, fmt='d')
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")

plt.show()
```



In [24]:
```
from sklearn.metrics import f1_score
```

In [25]:
```
accuracy = accuracy_score(ytest, y_pred)
error_rate = 1 - accuracy
precision = precision_score(ytest, y_pred)
recall = recall_score(ytest, y_pred)
f1 = f1_score(ytest, y_pred)
```

In [27]:
```
print(f"Accuracy: {accuracy:.2f}")
print(f"Error Rate: {error_rate:.2f}")
print(f"Precision: {precision:.2f}")
print(f"Recall: {recall:.2f}")
print(f"F1 Score: {f1:.2f}")
```

```
Accuracy: 0.89
Error Rate: 0.11
Precision: 0.89
Recall: 0.75
F1 Score: 0.81
```

In [ ]: